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Title: Allergic Bronchopulmonary Aspergillosis (ABPA) in Cystic Fibrosis Management

Introduction

ABPA a lung disease of hypersensitivity to *Aspergillus fumigatus* that occurs in asthmatic patients and cystic fibrosis.

The diagnosis of APBA in cystic fibrosis is difficult and often delayed because of shared clinical features with cystic fibrosis.

ABPA associated with poor lung function and progressive lung damage and fibrosis, so early diagnosis and treatment is essential

Diagnostic criteria:

Clinical:

Asthma symptoms, wheeze, dyspnea, cough and increased sputum production.

Acute/subacute Clinical deterioration

- Poor response to intravenous antibiotics in chest exacerbation.
- Decline in Pulmonary function (FEV1)

Radiology:

New chest radiography changes

Bronchiectasis (central) on CT chest

Serology:

- Total serum immunoglobulin E (IgE) greater than 500 international units (IU) per milliliter.
- High specific aspergillus IgERast. The normal value <0.35 IU/ml
- Positive skin test to an *Aspergillus* species
- Positive laboratory test (precipitin to *Aspergillus fumigatus* [*A. fumigatus*] or immunoglobulin G [IgG] antibody against *Aspergillus fumigatus* antigen)
- Blood work showing an elevation in eosinophilia>0.4x10⁹/l

1-Clinical deterioration, Asthma symptoms, decline in pulmonary function, increase in sputum production.

2-Total serum IgE >500 IU per ml. Should be done once patient off steroid.

3-Immediate cutaneous reactivity to Aspergillus OR strongly positive aspergillus Rast

4-Positive aspergillus precipitans OR new abnormalities on

Chest x-ray or CT scan not responding to standard therapy.

Treatment:

First line therapy: mainstay treatment of ABPA is oral steroid and antifungal therapy.

1-Oral corticosteroids

Prednisolone: 2mg/kg/day (maximum 60 mg/day) for 2weeks then 1mg/kg/day for 2 weeks then 1mg/kg/alternate day for 2weeks. Steroid dose is gradually tapered over 2-3 months.

*Precaution:

Bone prophylaxis, surveillance of glucose level, adjustment of insulin dose if applicable.

#Relapse is common within 2-3 years of 1st episode, and often high doses of steroids are needed for a long time.

2-Itraconazole:

Commence with steroid and continue for a minimum of 6 weeks and can be continued along with steroid course. better absorbed on empty stomach.

Dose 5mg/kg/day (maximum 400 mg/day).

If dose <200 mg/day given once daily

If dose >200mg/day given a twice daily

Duration 3-6 months

Second line therapy

3-Voriconazole:

Better absorption than itraconazole and not affected by gastric pH.

For patients who not responding or cannot tolerate itraconazole.

Dose: children <12 years 200 mg twice daily

Children >12 yrs. & <40 kg load with 200 mg twice daily for one day and then continue with 100 mg twice daily.

Children >12 yrs. >40 kg load with 400. Mg twice daily for one day then 200 mg twice daily.

Liver function tests (LFT's):

Should be performed regularly, usually at the same time as ABPA serology. Anti-fungal therapy, although not contraindicated in CF patients with established liver disease, caution should be exercised with this group of patients with LFT's is being closely monitored.

Drug interactions: itraconazole absorption is adversely affected by the concomitant use of antacids, H2 antagonists and proton pump inhibitors, and these should be stopped if possible. It can also cause increased systemic levels of antihistamines, with risks of ventricular arrhythmia.

May be considered in difficult cases at doses 5-10 mg twice daily after physiotherapy (check for bronchoconstriction and use bronchodilator pre- dose).

- Dissolve in water to achieve a concentration of 2.5mg/ml (20ml water in 50 mg amphotericin).
- The liposomal form would be considered only for those in whom bronchoconstriction is a severe problem

5-Intravenous pulse glucocorticosteroids:

In the literature, there are no randomized controlled trials comparing oral steroids to intravenous steroids. There are several cases reported the effectiveness of IV methylprednisolone in ABPA.

*Malena Cohen-Cymbarknoh 1 , Hannah Blau, David Shoseyov, Meir Mei-Zahav, Ori Efrati, Shoshana Armoni, Eitan Kerem.

Intravenous Monthly Pulse Methylprednisolone

Treatment for ABPA in Patients with Cystic Fibrosis. j cystic fibrosis .2009 Jul

6-Omalizumab (anti-IgE therapy) for allergic bronchopulmonary aspergillosis:

There is lack of evidence for the efficacy and safety of anti-IgE therapy in CF with ABPA. There are reported cases but there is a need for large prospective randomized controlled studies.

The Cochrane Database of Systematic Reviews, 18 Mar 2018, 3:CD010288

Anti-IgE therapy for allergic bronchopulmonary aspergillosis in people with cystic fibrosis.

Jet KR¹, Walia DK, Khairwa A

Monitoring of treatment efficacy: At 2-4 weeks since therapy started then 2-3 months

Evaluation of symptoms

Evaluation respiratory function FEV1

Follow-up chest imaging

Measuring serum IgE

References:

1-Allergic Bronchopulmonary Aspergillosis clinical care guidelines. Cystic fibrosis foundation. Reviewed in April 2019.

2-Standards for the Clinical Care of Children and a change of treatment.

Adults with cystic fibrosis in the UK December 2011.Cystic fibrosis trust

3-Norfolk, Suffolk & Cambridge shire Pediatric CF Network.

Cystic Fibrosis - Diagnosis and treatment of ABPA in children with cystic fibrosis October 2018

4-Malena Cohen-Cymbarknoh 1 , Hannah Blau, David Shoseyov, Meir Mei-Zahav, Ori Efrati, Shoshana Armoni, Eitan Kerem .

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5-Anti-IgE therapy for allergic bronchopulmonary aspergillosis in people with cystic fibrosis.

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